Chapter 1

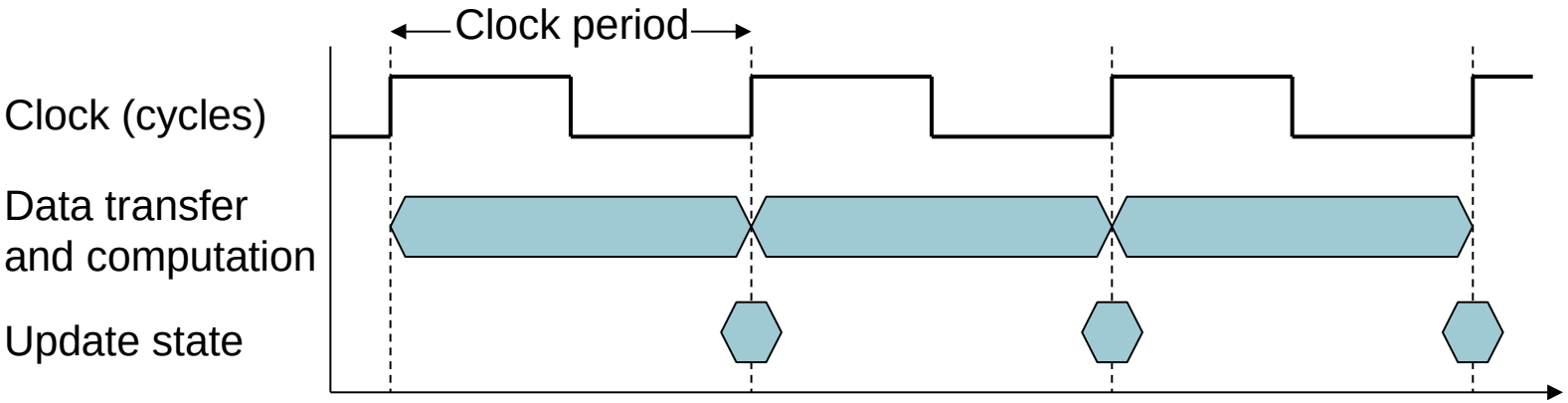
Computer Abstractions and Technology

Measuring Execution Time

- Elapsed time
  - Total response time, including all aspects
    - Processing, I/O, OS overhead, idle time
  - Determines system performance
- CPU time
  - Time spent processing a given job
    - Discounts I/O time, other jobs' shares
  - Comprises user CPU time and system CPU time
  - Different programs are affected differently by CPU and system performance

CPU Clocking

- Operation of digital hardware governed by a constant-rate clock



  - Clock period: duration of a clock cycle
    - e.g., 250ps = 0.25ns = 250×10^{-12}s

Q1. A processor operates at a clock rate of 2.5 GHz. What is its clock cycle time in nanoseconds (ns)?

A1. 0.4 ns

CPU Time

$$\text{CPU Time} = \text{CPU Clock Cycles} \times \text{Clock Cycle Time}$$

$$= \frac{\text{CPU Clock Cycles}}{\text{Clock Rate}}$$

- Performance improved by
  - Reducing number of clock cycles
  - Increasing clock rate
  - Hardware designer must often trade off clock

Q2. A program requires 8 x 10^9 clock cycles to execute on a 4 GHz processor. What is the CPU execution time?

A2. 2s

CPU Time Example

- Computer A: 2GHz clock, 10s CPU time

Q3. Match each term with its correct definition.

Term	Definition
1. Clock Cycle	A. The duration of one clock cycle
2. Clock Cycle Time	B. One basic timing interval of CPU op eration
3. Clock Rate	C. The number of clock cycles per sec ond

A3. B, A, C

$$= 10s \times 2GHz = 20 \times 10$$

$$\text{Clock Rate}_B = \frac{1.2 \times 20 \times 10^9}{6s} = \frac{24 \times 10^9}{6s} = 4GHz$$

Instruction Count and CPI

$$\text{Clock Cycles} = \text{Instruction Count} \times \text{Cycles per Instruction}$$

$$\text{CPU Time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock Cycle Time}$$

Q4. Which combination directly determines CPU execution time?

A. Instruction Count, CPI, and Clock Cycle Time

B. Instruction Count, Clock Rate, and Memory Size

C. CPI, Cache Size, and Clock Cycle Time

D. Clock Rate, Memory Size, and Instruction Size

A4. A

- If different instructions have different CPI
  - Average CPI affected by instruction mix

CPI Example

- Computer A: Cycle Time = 250ps, CPI = 2.0
- Computer B: Cycle Time = 500ps, CPI = 1.2
- Same ISA
- Which is faster, and by how much?

$$\text{CPU Time}_A = \text{Instruction Count} \times \text{CPI}_A \times \text{Cycle Time}_A$$

Q5. If CPU Time already measures how long a program takes to execute, why do we still need CPI?

A5. CPU Time tells us the overall performance, while CPI helps explain where the execution time comes from.

How fast → Why is it fast or slow?

CPI in More Detail

- If different instruction classes take different numbers of cycles

$$\text{Clock Cycles} = \sum_{i=1}^{n} (\text{CPI}_i \times \text{Instruction Count}_i)$$

Q6. A program executes three instruction classes. The relative frequencies of Class A and Class B are 40% and 35%, respectively. If the program executes 20,000 instructions in total, how many Class C instructions are executed?

A6.

$$f_A + f_B + f_C = 1, \qquad f_c = 1 - 0.40 - 0.35 = 0.25$$

$$IC_C = f_C \times IC$$

$$IC_C = 0.25 \times 20000 = 5000$$

CPI Example

- Alternative compiled code sequences using instructions in classes A, B, C

Class	A	B	C
CPI for class	1	2	3
IC in sequence 1	2	1	2
IC in sequence 2	4	1	1

CPI Example

- Alternative compiled code sequences using instructions in classes A, B, C

Class	A	B	C
CPI for class	1	2	3
IC in sequence 1	2	1	2
IC in sequence 2	4	1	1

- Sequence 1: IC = 5
  - Clock Cycles
    = 2×1 + 1×2 + 2×3
    = 10
  - Avg. CPI = 10/5 = 2.0

CPI Example

- Alternative compiled code sequences using instructions in classes A, B, C

Class	A	B	C
CPI for class	1	2	3
IC in sequence 1	2	1	2
IC in sequence 2	4	1	1

Q7. For Sequence 1 and Sequence 2, which instruction class contributes the most clock cycles?

A7. Sequence 1: Class C, Sequence 2: Class A

CPI Example

- Alternative compiled code sequences using instructions in classes A, B, C

Class	A	B	C
CPI for class	1	2	3
IC in sequence 1	2	1	2
IC in sequence 2	4	1	1

Q8. In Sequence 1, Class C contributes the most clock cycles. Does this also mean that Class C contributes the most to the CPU Time? Why?

A8. Yes, because in a same sequence, Clock Cycle Time is the same for all instruction classes

Q9. A program is modified in two ways:

Change 1: The algorithm is replaced by a more efficient algorithm that requires fewer operations.

Change 2: The program is rewritten in another programming language. After compilation, the new code uses more complex instructions that require more clock cycles on average

For IC, avg. CPI, Clock Cycle Time, determine whether it increases, decreases, or unchanged.

- Performance depends on

A9.

Performance Component	Change
Instruction Count (IC)	Decreases
Average CPI	Increases
Clock Cycle Time	Unchanged

- Instruction set architecture affects IC, CPI, T_c

Section 1-3 Quick Questions

Q1. A processor operates at a clock rate of 2.5 GHz. What is its clock cycle time in nanoseconds (ns)?

Q2. A program requires 8×10^9 clock cycles to execute on a 4 GHz processor. What is the CPU execution time?

Q3. Match each term with its correct definition: Clock Cycle, Clock Cycle Time, and Clock Rate.

Q4. Which combination directly determines CPU execution time?

Q5. If CPU Time already measures how long a program takes to execute, why do we still need CPI?

Section 1-3 Quick Questions

Q6. The relative frequencies of Class A and Class B are 40% and 35%. If the program executes 20,000 instructions in total, how many Class C instructions are executed?

Q7. For Sequence 1 and Sequence 2, which instruction class contributes the most clock cycles?

Q8. In Sequence 1, Class C contributes the most clock cycles. Does this also mean that Class C contributes the most to the CPU Time? Why?

Q9. A program is improved by using a more efficient algorithm, b ut the new compiled code uses more complex instructions. H ow do IC, average CPI, and Clock Cycle Time change?

Exercise 1 | Clock Rate and CPU Time

A processor operates at 3.2 GHz. A program requires 12.8 × 10⁹ clock cycles.

(a) Calculate the CPU execution time.
(b) If clock cycles are reduced by 25%, calculate the new CPU execution time.

Answer

(a) CPU Time = $(12.8 \times 10^9) / (3.2 \times 10^9) = 4$ s

(b) New Cycles = $0.75 \times 12.8 \times 10^9 = 9.6 \times 10^9$
New CPU Time = $(9.6 \times 10^9) / (3.2 \times 10^9) = 3$ s

Key Point: With a fixed Clock Rate, CPU Time changes in direct proportion to Clock Cycles.

Exercise 2 | Clock Rate vs. Clock Cycles

Computer A: CPU Time = 8 s, Clock Rate = 2.5 GHz.
Computer B: Clock Rate = 4.0 GHz, but requires 30% more cl ock cycles.

(a) Calculate CPU Time for Computer B.
(b) Calculate the speedup of B over A.

Answer

$\text{Cycles_A} = 8 \times 2.5 \times 10^9 = 20 \times 10^9$
$\text{Cycles_B} = 1.3 \times 20 \times 10^9 = 26 \times 10^9$

(a) CPU Time_B = 26 / 4 = 6.5 s
(b) Speedup = 8 / 6.5 ≈ 1.23×

Key Point: A higher Clock Rate does not guarantee the same perfo rmance gain if Clock Cycles also change.

Exercise 3 | Instruction Mix and Average CPI

Class A: CPI = 1, IC = 4×10^6
Class B: CPI = 2, IC = 3×10^6
Class C: CPI = 4, IC = 1×10^6 **Clock Rate = 2 GHz**

(a) Calculate average CPI. (b) Calculate CPU execution time.

Answer

Total IC = 8×10^6

(a) CPI_avg = [4(1) + 3(2) + 1(4)] / 8 = 1.75

(b) Clock Cycles = $8 \times 10^6 \times 1.75 = 14 \times 10^6$
CPU Time = $14 \times 10^6 / 2 \times 10^9 = 7$ ms

Key Point: Instruction Mix determines Average CPI, which affects Clock Cycles and CPU Time.

Exercise 4 | Compiler Optimization and Instruction Mix

Two compilers generate different instruction sequences for t he same program:

	Class A	Class B	Class C
CPI	1	2	5
Compiler X	6×10^6	2×10^6	2×10^6
Compiler Y	8×10^6	3×10^6	0.5×10^6

(a) Calculate total clock cycles for X and Y. (b) Which is faster?
(c) Why can Instruction Count alone be misleading?

Answer

(a) Cycles_X = $6(1)+2(2)+2(5) = 20 \times 10^6$
Cycles_Y = $8(1)+3(2)+0.5(5) = 16.5 \times 10^6$
(b) Compiler Y is faster.
(c) Y has more instructions, but far fewer high-CPI Class C instruc tions, so it needs fewer total cycles.

Key Point: More instructions can still run faster when the Instructio n Mix requires fewer total Clock Cycles.

Exercise 5 | Integrated Performance Analysis

Original: CPU Time = 1.6 s, IC = 2.0×10^9, CPI = 2.0, Clock Rate = 2.5 GHz.
A: IC ↓ 30%, CPI ↑ 10%. B: Clock Rate ↑ to 3.5 GHz.

(a) Calculate CPU Time after A. (b) Calculate CPU Time after B.
(c) Which optimization is better? Support with speedup.

Answer

(a) A: IC = 1.4×10^9, CPI = 2.2 → CPU Time = 1.232 s
(b) B: CPU Time = $(2.0 \times 10^9 \times 2.0) / (3.5 \times 10^9) \approx 1.143$ s

(c) Speedup_A ≈ 1.30×; Speedup_B ≈ 1.40×
Optimization B provides better performance.

Key Point: CPU Time must be evaluated using IC, CPI, and Clock Rate together.